Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/19/2015
PI Name:	Ritlop, Robert M.S.		
Project Title:	EyeBoxCNS		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10022	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI-RFA-SMARTCAP
Start Date:	05/01/2014	**End Date:**	04/30/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	9	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SMST03601		
Performance Goal No.:			
Performance Goal Text:			

When intracranial pressure (ICP) is increased, it can impact cranial nerves which can in turn reduce or alter the functional capabilities of these nerves, including the nerves required for eye movements. Thus by detecting functional changes in eye movement patterns—something easily done non-invasively—one could potentially identify elevated ICP. Eye tracking while watching a short film clip may be used to quantitate the physiologic impact of elevated intracranial pressure without requiring a trained examiner, pupil dilatation, imaging studies, or an invasive procedure. This represents a new application of eye tracking technology and a new mechanism for assessment of elevated intracranial pressure that is non-invasive, automatable, and could potentially be performed remotely. The research project began with two specific aims. (1) To determine eye tracking measures associated with measured intracranial pressure (ICP) values in patients with brain injury and other lesions (e.g., hydrocephalus) causing elevated pressure admitted to the ICU for invasive

Task Description:	neuromonitoring. (2) To demonstrate that subjects with mass effect (swelling) causing narrowing of the space adjacent to the third cranial nerve (perimesencephalic cistern or PMC) as assessed by volumetric analysis of CT (computed tomograph) scan imaging have decreased vertical eye movement on tracking. Our work demonstrated two major findings. The first is that eye tracking appears to be a viable approach for detecting the physiologic impact of elevated ICP. The second finding is that some patients demonstrate physiologic evidence of elevated ICP at values which are lower than those previously thought to represent thresholds for elevation. For example while 15 mm Hg is widely held to represent the value requiring treatment, we show that some patients are symptomatic at levels as low as 7 or 8. This finding is consistent with healthy subjects human data which suggests that normal healthy subject ICP values are generally less than 5 mm Hg. Our data further suggests that cellular microenvironment or brain tissue oxygenation may also impact physiologic cranial nerve function. We are planning studies to investigate the role of oxygenation in conjunction with ICP on physiologic function.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Oculogica is submitting two manuscripts with pilot data supporting the use of eye tracking to detect hydrocephalus and elevated ICP. These manuscripts will provide the basis for securing additional research funding for definitive Earth studies establishing sensitivity and specificity of eye tracking technology for elevated ICP. Simultaneously an FDA (Food and Drug Administration) submission is being compiled to enable commercial sale of our device. Major Oculogica accomplishments in the past year are numerous. Between August 2014 and January 2015, four provisional patent applications were generated. In chronological order of filing, these provisionals included assessment of the impact of morphine analogues on eye movement tracking, the ability to localize lesions of the medial longtitudinal fasciculus causing internuclear opthalmoplegia, and to differentiate it from post-nuclear nerve palsies with eye movement tracking, the impact of attention deficit hyperactivity disorder, chronic traumatic encephalopathy and schizophrenia spectrum disorders on eye movement tracking and finally the impact of ethanol consumption on eye movement tracking. The company's first utility patent was also pursued in Australia, Brazil, Canada, Europe, Japan, and the US. Original research published relevant to Oculogica's eye tracking technology base includes the cranial nerve palsy paper published January 2015 (A Novel Method for Eye Tracking While Watching a Short Film Clip Detects Cranial Nerve III and VI Palsies) and the concussion paper published soon thereafter (Eye tracking detects disconjugate eye movements associated with structural traumatic brain injury and concussion). A third paper introducing eye tracking as a biomarker for brain injury is also accepted for publication with several more in preparation, including the elevated intracranial pressure paper based on this study. All of the progress Oculogica is making is intended to support an application to the FDA to market a medical device commercially to Earth markets. The commercial endeavor is centered on concussion due to several mechanistic underlying pathologies, including intracranial pressure. Oculogica is targeting a submission in the first quarter of 2016.
Task Progress:	Oculogica's project was based in Bellevue Hospital neurosurgical ICU in Manhattan, New York. In total, 24 patients (12 females, 12 males) were admitted to the neurosurgical intensive care unit for intracranial pressure monitoring and were enrolled in the study (mean age=47; range 26 to 65 years old). 19 patients presented with hemorrhage (15 aneurysmal subarachnoid hemorrhage (SAH), 3 traumatic SAH, and 1 intraventricular hemorrhage (IVH)) and 5 with obstructive hydrocephalus (4 intra-axial tumors, 1 bilateral cerebellar infarcts). These 24 patients were eye-tracked in 78 separate sessions: 59 times with either intracranial pressure monitors or external ventriculostomy drains in place and actively monitoring ICP, and 19 times after discontinuation of monitoring or replacement of catheters by a ventriculoperitoneal shunt. Of the 24 patients, 18 were tracked serially at different ICPs. One 64-year-old male, who was eye-tracked once, was excluded after enrollment in this study due to insufficient ability to maintain attention while watching the stimulus. The remaining 5 were either discharged, had an additional operation for placement of a ventriculoperitoneal shunt, or expired. Spearman correlation analysis of eye tracking metrics in patients with ICP monitors in place demonstrates decreased box width and area for the right and left eyes, respectively, and decreased horizontal conjugacy with elevated ICP. Analysis of individual patient examples showed that as ICP increased, these metrics worsened and as it decreased, the values returned towards normal.
Bibliography Type:	Description: (Last Updated: 10/20/2015)
Articles in Peer-reviewed Journals	Samadani U, Kolecki R, Chaudhry O, Ritlop R, Mehmood T, Han BJ, Reyes M, Kondziolka D, Huang P. "Elevated Intracranial Pressure Causes Reversible Alterations in Eye Tracking Detectable While Watching a Short Film Clip." Submitted, June 2014. Status as of October 2015--not yet published or in press. , Jun-2014